

Livinggo — Complete Product Requirements Document (PRD)

PG Accommodation Marketplace & Management Platform

Version 1.0 — Founder Specification Draft

1. PROJECT OVERVIEW

Product Name

Public Brand:

Livinggo

Product Type

A full-stack PG accommodation discovery, booking, brokerage protection, tenant management, and operations platform.

The platform will include:

1. Student-facing portal/app
2. PG owner portal/dashboard
3. Internal staff/admin dashboard
4. Super admin master control panel
5. Unified backend/database
6. Mobile-first architecture
7. Progressive Web App compatibility
8. Future Android/iOS conversion support

2. CORE BUSINESS OBJECTIVE

The platform's primary objective is:

A. Student Acquisition

Allow students to discover and book PG accommodations.

B. Owner Management

Allow PG owners to list and manage properties.

C. Brokerage Protection (MOST IMPORTANT)

The system **MUST** prevent or strongly discourage bypassing LivingCo.

This is the highest priority business logic.

There should **NEVER** be a scenario where:

- Student discovers PG via Livinggo
- Owner and student connect directly
- Livinggo loses brokerage/revenue

The system architecture must be designed specifically to minimize this.

3. PRIMARY USER ROLES

3.1 Student Role

Permissions

Students can:

- Create account
- Browse PGs
- View PG interiors
- Book PG

- Make payments
- Raise complaints
- View rent history
- Chat with support
- Access food menu/notices
- View agreements/documents
- Request room changes

Restrictions

Students CANNOT:

- Directly contact owner initially
- View exact building exterior
- View exact property coordinates before booking
- Access owner phone number before booking confirmation
- Download sensitive property information

3.2 PG Owner Role

Permissions

Owners can:

- Add/manage PG listings
- Upload room images
- Upload interior-only virtual tours
- View leads
- Accept/reject applications
- Manage occupancy
- View rent status
- View complaints
- Manage notices/menu

Restrictions

Owners CANNOT:

- Access student personal contact before controlled unlock
- Share phone number publicly
- Edit brokerage policies
- Remove LivingCo references
- Circumvent platform agreements
- Download full student databases

3.3 Staff/Admin Role

Permissions

Staff admins can:

- Approve listings
- Moderate complaints
- Verify KYC
- Review uploaded documents
- Handle support tickets
- Monitor transactions
- Monitor suspicious activity

Restrictions

Staff admins CANNOT:

- Delete super admin
- Change platform policies
- Access full financial controls
- Modify commission structures
- Access master credentials
- Permanently delete properties

3.4 Super Admin Role (Founder-Level Access)

Full unrestricted platform control

Super Admin can:

- Add/remove admins
- Suspend owners/students
- Revoke listings
- Edit platform fees
- Override bookings
- Access analytics
- Access financial reports
- View suspicious activity
- Control all permissions
- Control all APIs
- Access audit logs
- Reset accounts
- Modify policies
- View brokerage recovery data

Important Security Requirement

Super Admin access must be:

- Email-linked
- MFA protected
- IP monitored
- Device authenticated
- Logged in audit trails

4. ANTI-BYPASS SYSTEM (MOST CRITICAL SECTION)

PRIMARY BUSINESS PROBLEM

Without protections:

- Student discovers PG
- Student visits property
- Owner offers direct discount
- LivingCo loses brokerage

This must be prevented technically + operationally + contractually.

5. REQUIRED ANTI-BYPASS SYSTEMS

5.1 Contact Masking System

Before booking/payment:

Student MUST NOT see:

- Owner phone number
- Exact address
- Full building details
- WhatsApp link

Owner MUST NOT see:

- Student personal number
- Personal email

All communication should happen through:

- In-app chat
- Masked calling
- Temporary relay numbers (future phase)

5.2 Property Address Obfuscation

Before payment:

Display only:

- Area/locality
- Landmark radius
- Approximate map zone

DO NOT display:

- Exact building
- Flat number
- Exterior visuals

5.3 Interior-Only 3D PG Tours

VERY IMPORTANT REQUIREMENT

3D tours **MUST ONLY** show:

- Room interiors
- Hallways
- Washrooms
- Kitchens
- Amenities

STRICTLY PROHIBITED:

- Building exterior
- Street view
- Balcony external angle
- Signboards
- Nearby shops
- Geolocation clues

Purpose:
Prevent direct identification of PG.

5.4 Lead Locking System

Once a student views/bookmarks/applies for a PG:

The lead becomes “platform tagged”.

Meaning:

- Owner cannot directly close outside deal
- System records:
 - timestamp
 - device
 - visit history
 - interaction logs

5.5 Mandatory Booking Token

Student must pay:

“LivingGo Booking Token”

Before:

- Physical visit
- Owner details unlock
- Final booking

This token:

- Non-refundable OR partially refundable
- Adjusted in booking amount

This psychologically commits student into platform flow.

5.6 Platform-Managed Visit Scheduling

Visits **MUST** happen only through:

- Scheduled slots
- QR verification
- Visit pass
- In-app approval

Owner cannot manually invite random student.

5.7 Anti-Circumvention Agreement

Both owner and student must digitally agree:

Student Clause

If booking occurs outside platform after lead generation:

- Token forfeited
- Blacklisting possible

Owner Clause

If bypass detected:

- Penalty fee
- Listing suspension
- Permanent removal
- Security deposit forfeiture

5.8 Delayed Contact Reveal

Owner/student contact details revealed ONLY after:

- Booking token payment
- Visit confirmation
- Platform acknowledgment

5.9 Hidden Watermarking

Images/tours uploaded should contain:

- Invisible watermarking

- Metadata tagging

To detect external reposting.

5.10 AI Suspicion Detection (Future Phase)

Flag suspicious patterns:

- Student viewing same PG repeatedly
- Sudden inactive lead after visit
- Owner not responding inside platform
- External negotiation patterns

5.11 Visit OTP Verification

When student physically visits:

- Owner enters OTP generated by app
- Visit officially logged

Without OTP:

- Visit considered invalid

This creates traceability.

5.12 Commission Recovery Tracking

If owner rents to student later:

- Owner legally liable for commission

System should maintain:

- lead history
- timestamps
- proof of introduction

6. MOBILE-FIRST REQUIREMENT

CRITICAL DEVELOPMENT REQUIREMENT

Entire system MUST be built:

- Mobile responsive
- PWA compatible
- App-convertible

Technology Requirement

Frontend must support:

- Android conversion
- iOS conversion
- Same codebase

Avoid:

- Separate mobile development later

Recommended:

React + Next.js + Tailwind + Firebase/Supabase.

7. STUDENT APP FEATURES

Dashboard

- Booking status
- Rent due
- Notices
- Food menu
- Complaints
- Support

PG Discovery

- Search

- Filters
- Budget
- Distance
- Amenities
- Ratings

PG Detail Page

- Interior images
- 3D tour
- Room availability
- Rules
- Amenities
- Approximate location only

Booking System

- Booking token payment
- Visit scheduling
- Agreement acceptance

Payments

- Rent
- Deposit
- Booking fees
- Utility bills

Complaint System

- Raise complaint
- Upload images

- Track status

Digital Agreements

- E-sign support
- PDF storage

Notifications

- Push notifications
- Email
- WhatsApp alerts

8. OWNER PANEL FEATURES

Dashboard

- Occupancy
- Revenue
- Leads
- Complaints
- Payments

Property Management

- Add/edit property
- Add rooms/beds
- Pricing
- Availability

Lead Management

- Applications

- Visit requests
- Booking requests

Financial Management

- Rent status
- Revenue reports
- Brokerage deductions

Complaint Management

- Resolve complaints
- Update status

Notices/Menu

- Publish food menu
- Announcements

9. SUPER ADMIN PANEL (MASTER CONTROL PANEL)

MOST IMPORTANT BACKEND PANEL

Must have full system control.

Modules Required

User Management

- Add/remove users
- Suspend accounts

Property Control

- Approve/reject PGs

- Revoke listings

Financial Control

- Brokerage tracking
- Revenue reports

Complaint Oversight

- Escalations
- Resolution audits

Analytics

- Occupancy
- Revenue
- Conversion rate
- Bypass detection

KYC Verification

- Aadhaar/PAN review
- Manual approval

Access Control

- Role management

Audit Logs

Track EVERY:

- edit
- login
- deletion
- policy change

10. SECURITY REQUIREMENTS

Mandatory Security Features

- HTTPS only
- JWT auth
- MFA for admins
- Role-based access control
- Rate limiting
- Encrypted passwords
- Audit logging
- Secure file uploads
- Database rules

11. PAYMENT SYSTEM

Integrate:

Preferred:

- Razorpay

Support:

- UPI
- Cards
- Netbanking

12. DATABASE MODULES

Collections/Tables

- users
- students
- owners
- staff

- properties
- rooms
- bookings
- visits
- payments
- complaints
- menus
- notices
- agreements
- analytics
- audit_logs
- suspicious_activity

13. FUTURE FEATURES

AI Recommendation Engine

Recommend PGs.

Dynamic Pricing

Demand-based pricing.

Smart Occupancy Prediction

AI Chatbot

Referral System

Subscription Plans For Owners

14. RECOMMENDED TECH STACK

Frontend

- React / Next.js
- Tailwind CSS

Backend

- Node.js / Express

Database

- PostgreSQL OR Firebase

Hosting

- Vercel
- AWS

Storage

- Cloudinary/S3

Authentication

- Firebase Auth/Auth0

15. FINAL DEVELOPMENT REQUIREMENTS

Developer must ensure:

- Scalable architecture
- Clean modular code
- Mobile-first development
- API documentation
- Admin documentation

- Error logging
- Analytics integration
- SEO optimization
- Fast loading
- App-ready structure

16. FINAL FOUNDER NOTE

The platform's success depends heavily on:

1. Preventing owner-student bypass
2. Controlling lead flow
3. Restricting direct contact
4. Keeping transactions platform-controlled
5. Protecting brokerage economics

Every product decision should prioritize these objectives.

Entire “**Anti-Circumvention & Brokerage Protection Framework**” section to the PRD.

Add the following sections to the document.

17. LEGAL & CONTRACTUAL ENFORCEMENT SYSTEM

17.1 Mandatory Digital Acceptance System

Before using the platform:

Students MUST digitally accept:

- Terms & Conditions
- Anti-circumvention policy
- Brokerage acknowledgment

- Lead introduction acknowledgment

PG Owners MUST digitally accept:

- Platform listing agreement
- Brokerage agreement
- Non-circumvention agreement
- Commission recovery clause
- Lead ownership clause

Acceptance must include:

- Timestamp
- IP logging
- Device logging
- Browser fingerprinting
- User ID linkage

This data must be stored permanently for legal evidence.

17.2 Lead Ownership Clause

IMPORTANT LEGAL PRINCIPLE

Any student introduced to a PG owner through Livingo shall legally constitute a “platform-generated lead”.

This applies if:

- Student viewed listing
- Student booked visit
- Student contacted through platform
- Student paid token
- Student interacted with owner

Once tagged as a Livingo-generated lead:

Owner cannot:

- Complete transaction outside platform
- Offer off-platform discounts
- Avoid brokerage payment
- Encourage bypass

for a defined protection period.

17.3 Lead Protection Period

Mandatory Clause

Every lead generated through Livingo shall remain protected for:

Recommended:

- 12 months from first interaction

OR

- 6 months minimum

Even if:

- student books later,
- student contacts later,
- student shifts later,
- transaction happens offline later.

Brokerage liability survives.

17.4 Non-Circumvention Clause (OWNER)

Mandatory Owner Agreement Clause

Suggested wording:

“The Owner agrees not to directly or indirectly bypass Livingo for any tenant, lead, inquiry, or booking introduced through the Livingo platform. Any attempt to complete an off-platform transaction with a Livingo-generated lead shall constitute a material breach of agreement.”

17.5 Brokerage Recovery Clause

Suggested clause:

“In the event of circumvention, the Owner shall remain liable to pay the full brokerage/commission amount due to Livingo, along with any legal costs, recovery costs, penalties, and damages arising from such breach.”

17.6 Platform Introduction as Valid Consideration

Add clause:

“The introduction of tenant and property through Livingo constitutes valuable commercial consideration and shall be sufficient basis for commission entitlement.”

This is important legally.

17.7 Digital Evidence Preservation System

Platform must preserve:

- chat logs
- visit logs
- OTP verification
- timestamps
- booking attempts
- token payments
- owner responses
- property interactions
- IP/device data

These records should be exportable as PDFs for legal proceedings.

17.8 Visit Verification Record

Every physical visit must generate:

- Visit ID
- OTP verification
- Date/time
- Student identity
- Owner acknowledgment

This becomes evidence of lead introduction.

17.9 Anti-Bypass Detection System

System should automatically flag suspicious scenarios:

Examples

Owner side:

- Lead suddenly becomes inactive after visit
- Owner stops responding
- Multiple repeated visits without booking

Student side:

- Repeated property revisits
- Contact exchange attempts in chat

Platform response:

- Trigger compliance alerts
- Trigger support intervention

17.10 In-App Chat Monitoring Rules

System should automatically block:

- phone numbers
- WhatsApp numbers
- email IDs

- Telegram IDs
- external links

before booking confirmation.

Example:

If owner types:

“Call me at 98XXXXXX”

System masks message automatically.

17.11 Hidden Contact Protection

Before booking confirmation:

Student should NOT receive:

- owner phone number
- exact address
- direct WhatsApp

Owner should NOT receive:

- student number
- personal email

All communication routed through Livingo.

17.12 Mandatory Booking Token

Student must pay a booking/visit token before:

- exact address reveal
- owner contact reveal
- visit confirmation

Purpose:

Increase commitment and reduce bypass probability.

17.13 Penalty Clause (Owner)

Suggested clause:

“Any owner found attempting to circumvent Livingo by engaging in direct/off-platform transactions with platform-generated leads shall be liable for:

- Full brokerage amount
- Additional penalty charges
- Listing suspension/removal
- Legal recovery proceedings
- Blacklisting from platform”

17.14 Student Anti-Circumvention Clause

Suggested clause:

“Students agree not to intentionally bypass Livingo after being introduced to a property through the platform. Violation may result in forfeiture of booking token, account suspension, and legal action where applicable.”

17.15 Jurisdiction Clause

Add:

“All disputes shall be subject to the exclusive jurisdiction of courts located in [Your Preferred City].”

Recommended:

Delhi NCR if company registered there.

17.16 Arbitration Clause

Add arbitration mechanism for faster recovery.

Example:

“Disputes may first be resolved through arbitration before formal litigation.”

17.17 E-Sign & Legal Acceptance

Every major action should require:

- checkbox acceptance
- digitally stored consent
- timestamp

Applicable for:

- owner onboarding
- booking
- token payment
- agreements

17.18 Owner Security Deposit (VERY IMPORTANT)

Strong recommendation:

Charge owners:

- refundable onboarding/security deposit

Example:

₹5,000–₹25,000.

If bypass proven:

- adjusted against penalties.

This is psychologically extremely effective.

17.19 Legal Notice Automation

Admin panel should include:

“Generate Legal Notice” Feature

System auto-generates:

- owner details
- lead evidence
- timestamps

- breach evidence

Export:

- PDF
- advocate-ready notice draft

17.20 Audit Trail System

Every action must be logged:

- edits
- lead access
- visit approval
- booking activity
- cancellations

Logs should NOT be deletable except by super admin.

17.21 Data Retention Policy

All lead and transaction data should be retained for:

- minimum 5 years

for:

- legal recovery
- taxation
- compliance
- disputes

17.22 Founder Recommendation (IMPORTANT)

Even the best technical system cannot stop bypass completely.

The actual winning formula is:

COMBINATION OF:

- contact masking
- tokenization
- delayed reveal
- legal liability
- visit logging
- security deposits
- platform convenience
- trust ecosystem

Together they make bypass:

- risky,
- inconvenient,
- and legally expensive.

That is the real objective.